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Multimodal radiopathomics model predicts postoperative metachronous liver metastasis in gastric cancer

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Ping'an Ding[1,2,3,17], Jiaxuan Yang[1,2,3,17], Honghai Guo[1,2,3], Sheng Chen[1,2,3], Yueping Liu[4], Xiao Han[5], Songpu Li[6,7], Jiaxiang Wu[1,2,3], Li Yang[8], Haotian Wu[1,2,3], Ning Meng[9], Xiaolong Li[10], Renjun Gu[11,12], Yuan Tian[1,2,3], Lilong Zhang[13], Zhenjiang Guo[14], Le Lu[15], Sen Yang[15], Lingjiao Meng[16] & Qun Zhao[1,2,3]

In locally advanced gastric cancer, a substantial number of patients relapse with liver metastasis months after an apparently curative operation, and standard tumor staging offers little warning of who is at risk. Here, we develop the Radiopathomics-Clinical Stratification Assessment (RCSA), an interpretable model that integrates three complementary sources of information: radiomic features from preoperative computed tomography, pathomic features from routine hematoxylin and eosin tumor slides, and conventional clinical features. Trained on patients from one hospital and then tested on separate internal, external, public, and prospective trial groups (NCT02555358), RCSA consistently separates high- and low-risk patients, with area under the curves between 0.862 and 0.909. Tumors it labels low-risk carry a notably more active immune environment, indicating that these patients are the ones most likely to gain from added immunotherapy. RCSA therefore turns existing hospital data into individualized guidance for postoperative follow-up and treatment.

Gastric cancer (GC) remains one of the leading causes of cancer-related deaths worldwide, particularly in East Asian countries, including China, Japan, and South Korea[1,2]. Despite improvements in surgical techniques, chemotherapy regimens, and targeted therapies, the prognosis for patients with locally advanced gastric cancer (LAGC) remains suboptimal due to the high rate of distant metastases, particularly metachronous liver metastasis (MLM) occurring after curative resection[3,4]. MLM has been consistently associated with poor clinical

[1]The Third Department of Surgery, the Fourth Hospital of Hebei Medical University, Shijiazhuang, Hebei, China. [2]Hebei Key Laboratory of Precision Diagnosis and Comprehensive Treatment of Gastric Cancer, the Fourth Hospital of Hebei Medical University, Shijiazhuang, Hebei, China. [3]Hebei Provincial Engineering Research Center for Big Data Analysis and Mining Application for Precise Diagnosis and Treatment of Gastric Cancer, the Fourth Hospital of Hebei Medical University, Shijiazhuang, Hebei, China. [4]Department of Pathology, The Fourth Hospital of Hebei Medical University, Shijiazhuang, Hebei, China. [5]College of Biomedical Engineering, Sichuan University, Chengdu, Sichuan, China. [6]Intelligent Information Technology Research Center, Jingchu University of Techology, Jingmen, China. [7]Jingmen Cryptometry Application Techology Research Center, Jingmen, China. [8]The Department of CT/MRI, The Fourth Hospital of Hebei Medical University, Shijiazhuang, Hebei, China. [9]Department of General Surgery, Shijiazhuang People's Hospital, Shijiazhuang, Hebei, China. [10]Department of General Surgery, Baoding Central Hospital, Baoding, Hebei, China. [11]School of Chinese Medicine & School of Integrated Chinese and Western Medicine, Nanjing University of Chinese Medicine, Nanjing, Jiangsu, China. [12]Department of Gastroenterology and Hepatology, Jinling Hospital, Medical School of Nanjing University, Nanjing, Jiangsu, China. [13]Department of General Surgery, Renmin Hospital of Wuhan University, Wuhan, Hubei, China. [14]Department of General Surgery, Hengshui People's Hospital, Hengshui, Hebei, China. [15]Ant Healthcare (AFU), Ant Group, Hangzhou, Zhejiang, China. [16]Research Center and Tumor Research Institute of the Fourth Hospital of Hebei Medical University, Shijiazhuang, Hebei, China. [17]These authors contributed equally: Ping'an Ding, Jiaxuan Yang. e-mail: ding_ping_an@hebmu.edu.cn; sen.yang.scu@gmail.com; ljmeng@hebmu.edu.cn; zhaoqun@hebmu.edu.cn

outcomes and significantly compromises long-term survival, highlighting the urgent need for reliable predictive biomarkers and risk stratification tools to guide postoperative management and individualized treatment strategies for these patients[5–7].

Previous efforts to predict postoperative recurrence and metastasis in GC have primarily relied on clinical and pathological characteristics, including tumor-node-metastasis (TNM) stage, tumor differentiation, Lauren classification, and serum tumor markers[8–10]. However, such clinical prognostic models often demonstrate limited accuracy and reproducibility due to the inherent biological and clinical heterogeneity of GC. Recently, radiomics, a quantitative imaging approach extracting high-dimensional features from standard medical images, has emerged as a promising predictive tool, capable of capturing tumor heterogeneity and underlying biological behavior beyond conventional imaging evaluation[11–13]. For instance, prior studies have demonstrated the potential of radiomic features derived from computed tomography (CT) scans to accurately predict recurrence and survival outcomes in gastric cancer patients, indicating the advantage of integrating radiological biomarkers into clinical practice[14–16].

Parallelly, digital pathology, particularly deep learning-based pathomics, has increasingly been recognized as a powerful modality to provide detailed insights into tumor microenvironment (TME) characteristics and histopathological heterogeneity[17,18]. Deep learning methods have facilitated the analysis of whole-slide images (WSI), capturing subtle histological patterns and complex spatial relationships between tumor cells and surrounding stromal or immune cells[19,20]. Recent studies have shown that deep learning-derived pathological features are significantly associated with clinical outcomes and treatment response in patients with gastric cancer[21–26]. However, despite these promising advances, most previous predictive models have been developed based on either radiomics or pathomics independently, and few studies have systematically integrated these complementary modalities with clinical predictors to build comprehensive multimodal prediction models.

Here, we develop and validate a clinically interpretable multimodal model, the Radiopathomics-Clinical Stratification Assessment (RCSA), which integrates conventional clinical variables, preoperative CT-derived radiomic features, and deep learning-based pathomic features from routine hematoxylin and eosin slides to predict postoperative MLM in patients with LAGC. Across a retrospective multicenter development cohort and multiple internal, external, public, and prospective trial validation cohorts, the RCSA model shows strong and stable discrimination (area under the curves ranging from 0.862 to 0.909) and outperforms clinical-only and single-modality models, while also stratifying patients by overall survival. Interpretability analysis identifies the imaging and pathological features that contribute most to risk. Transcriptomic sequencing and immune profiling further reveal that RCSA-defined low-risk tumors harbor an immune-enriched microenvironment with greater $CD8^+$ T cell infiltration and elevated immune-checkpoint gene expression, indicating that these patients may be more responsive to adjuvant immunotherapy. Together, these findings suggest that the RCSA model may support risk-adapted postoperative surveillance and personalized treatment in LAGC.

Results

Patient cohorts and clinical characteristics

A total of 1878 eligible patients with LAGC were included for model development and multi-level validation (Fig. 1). The overall cohort structure comprised a Training Cohort ($N = 770$), an Internal Validation Cohort ($N = 362$), an External Validation Cohort ($N = 489$, including two regional clinical cohorts and one public database cohort from TCIA), and a Prospective Validation Cohort (NCT02555358, $N = 257$). The clinicopathological characteristics were well balanced across the six participating medical centers, with a clear male predominance (65.6% male versus 34.4% female), consistent with the recognized sex distribution of gastric cancer (Supplementary Table 4).

The training cohort ($n = 770$) had a median follow-up of 69 months (95% CI, 67.5–71.0), during which 128 MLM events were observed (16.6%). The internal validation cohort ($n = 362$) was stratified by treatment period into an earlier cohort (cohort I; $n = 194$) and a later cohort (cohort II; $n = 168$). Median follow-up was 62 months (95% CI: 58.5–64.5) in cohort I, with 28 MLM events (14.4%), and 54 months (95% CI: 50.0–57.5) in cohort II, with 27 MLM events (16.1%).

The external validation cohort was subdivided by geographic region. The External Validation I ($n = 257$) had a median follow-up of 53.5 months (95% CI, 51.0–56.0), with 45 MLM events (17.5%), whereas the External Validation II ($n = 191$) had a median follow-up of 58.7 months (95% CI: 55.0–62.0), with 32 MLM events (16.8%). The External Validation III (TCIA cohort; $n = 41$) consisted of patients retrieved from a public database; in this cohort, 6 MLM events (14.6%) were identified. Due to the nature of the public dataset, long-term survival follow-up was not available for this specific group. The prospective validation cohort ($n = 257$) had a median follow-up of 63 months (95% CI, 60.5–65.5), with 38 MLM events observed (14.8%).

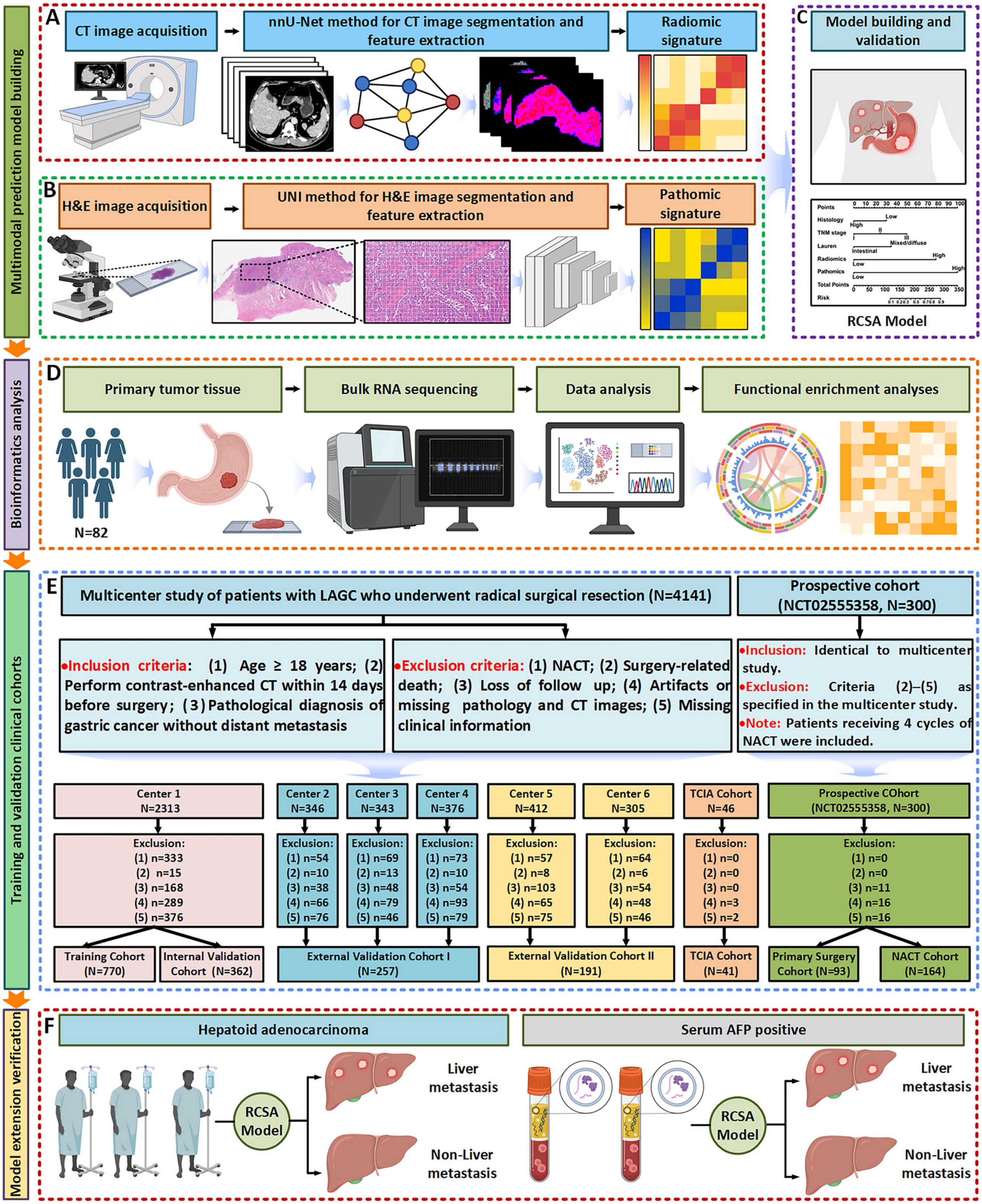
Development and validation of the RCSA model for predicting postoperative MLM

Multivariate logistic regression analysis identified pathological type (OR = 2.125, 95% CI: 1.226–3.685, $P = 0.007$), postoperative pTNM stage (OR = 3.688, 95% CI: 1.124–12.099, $P = 0.031$), Lauren classification (OR = 2.708, 95% CI: 1.571–4.669, $P = 0.001$), radiomic features (OR = 7.873, 95% CI: 4.651–13.328, $P < 0.001$), and pathomic features (OR = 13.474, 95% CI: 8.037–22.590, $P < 0.001$) as independent predictors of MLM in LAGC patients (Supplementary Table 5). Based on these variables, a comprehensive nomogram was constructed to develop the Radiopathomics-Clinical Stratification Assessment (RCSA) model (Fig. 2A), which demonstrated strong predictive performance (AUC = 0.909, 95% CI: 0.881–0.937; Fig. 2B–E).

Model comparison showed superior predictive performance of the RCSA model over models using clinical, radiomic, or pathomic features alone (Supplementary Table 6). We compared the performance of the RCSA model with a state-of-the-art transformer-based multimodal deep learning architecture. RCSA exceeded the performance of the transformer-based model, achieving AUC values ranging from 0.865 to 0.908 across the internal, external, and prospective validation cohorts, compared with 0.708–0.869 for the transformer-based model (Supplementary Fig. 3 and Supplementary Table 7). Compared with the clinical model, the RCSA model achieved significant improvements in prediction accuracy, with a NRI of 0.658 (95% CI: 0.467–0.850, $P < 0.001$) and an IDI of 0.376 (95% CI: 0.326–0.426, $P < 0.001$). DeLong's test further confirmed that the RCSA model significantly outperformed the clinical model (AUC: 0.909 vs. 0.701, $P < 0.001$; Supplementary Table 8), and its calibration curve supported strong predictive accuracy (Brier score = 0.080; Supplementary Fig. 4B).

In addition, sensitivity analyses were conducted using multiple imputation by chained equations (MICE) to account for missing clinical data. When re-evaluated on the imputed dataset, the RCSA model achieved an AUC of 0.861 (95% CI: 0.840–0.883), compared with an AUC of 0.895 (95% CI: 0.873–0.917) obtained from the complete-case analysis in the overall dataset (Supplementary Fig 5 and Supplementary Table 9).

Risk stratification based on the Youden index classified patients into low- and high-risk groups, with the RCSA model identifying more low-risk patients with MLM compared to the clinical model (70.3% vs. 60.8%; Supplementary Fig. 4D). Decision curve and clinical impact analyses further demonstrated the RCSA model's superior clinical utility (Fig. 2F and Supplementary Fig 4A).



Survival analysis revealed significantly lower 5-year OS in high-risk patients compared to low-risk patients (38.9% vs. 55.6%; $P < 0.001$; Fig. 2G). In multivariate Cox regression adjusted for clinicopathological factors (including N stage and the receipt of adjuvant chemotherapy), the RCSA model remained an independent prognostic factor for 5-year OS (HR = 1.517, 95% CI: 1.225–1.880; Supplementary Table 10). In addition, the RCSA model achieved a concordance index (C-index) of 0.660 (95% CI: 0.634–0.686) in the training set, indicating moderate discriminative performance for survival prediction (Supplementary Table 11).

Bootstrap resampling (1000 iterations) confirmed the RCSA model's superior diagnostic performance for MLM prediction (Mean AUC = 0.909, 95% CI: 0.881–0.935) compared to the clinical model (Mean AUC = 0.701, 95% CI: 0.654–0.749; Supplementary Fig. 4C). Ten-fold cross-validation further supported the model's robustness, demonstrating superior discrimination, sensitivity, and specificity

Fig. 1 | Study design overview. A Radiomics Workflow: After image acquisition, segmentation was performed using nnU-Net. Radiomic features were extracted and modeled using 3D Slicer. **B Pathomics Workflow:** Whole slide images (WSIs) were tiled into patches, which were input into a feature extractor to generate pathomic features. **C RCSA Model Development:** Construction and validation of the RCSA model for predicting MLM in patients with LAGC, along with performance evaluation. **D Bioinformatics Workflow:** RNA sequencing was conducted on 82 prospectively collected tissue samples. Bioinformatic analyses were performed to explore biological characteristics and immune infiltration patterns associated with key features, reflecting tumor heterogeneity in LAGC. **E Study Cohort:** A total of 1878 eligible patients with LAGC were enrolled from six medical centers in China and a public dataset (TCIA cohort). All patients had preoperative abdominal CT scans and postoperative H&E-stained pathological slides. The cohort structure includes training, internal, and external validation sets (predominantly primary surgery) and a final prospective validation set (including both primary surgery and NACT patients). **F Model Extension and Verification:** The predictive performance of the RCSA model was further verified in specialized populations, including hepatoid adenocarcinoma and serum AFP-positive gastric cancer, to assess its generalizability for liver metastasis prediction. **Participating Centers:** Center 1: The Fourth Hospital of Hebei Medical University; Center 2: Shijiazhuang People's Hospital; Center 3: Baoding Central Hospital; Center 4: Hengshui People's Hospital; Center 5: Jinling Hospital, Nanjing; Center 6: Renmin Hospital of Wuhan University. AFP alpha-fetoprotein; CT computed tomography; H&E haematoxylin and eosin; LAGC locally advanced gastric cancer; MLM metachronous liver metastasis; NACT neoadjuvant chemotherapy; nnU-Net, no-new-Net segmentation framework; RCSA Radiopathomics-Clinical Stratification Assessment; TCIA The Cancer Imaging Archive; TNM tumour–node–metastasis; UNI pathology foundation model used for pathomic feature extraction; WSI whole-slide image. Schematic illustrations in this figure were created with BioRender.com.

compared to single-feature models (Fig. 2H). Subgroup analyses stratified by serum tumor markers (CA72-4, CA19-9, CEA) and HER2 status also confirmed the RCSA model's superior predictive performance (Fig. 2I). Detailed results for these analyses are presented in the Supplementary Information (Supplementary Figs. 4E, 6–8 and Supplementary Tables 12–13). Additionally, stratification based on serum AFP status and the presence of Hepatocellular carcinoma (HCC) demonstrated that the RCSA model consistently maintained the highest AUC (ranging from 0.860 to 0.917) and overall superior performance compared to Pathomics, Radiomics, and Clinical models in all subgroups (Supplementary Fig 9**;** Supplementary Tables 12–13).

Internal validation of the RCSA model

To further validate the RCSA model, we retrospectively analyzed 362 LAGC patients treated at the FHHMU, divided into two internal validation cohorts based on treatment period: Cohort I (January 2012–December 2013, $n = 194$) and Cohort II (January 2018–December 2019, $n = 168$). Clinical characteristics were comparable between cohorts (all $P > 0.05$; Supplementary Table 14). In Cohort I, the RCSA model demonstrated superior predictive performance compared to models based solely on clinical, radiomic, or pathomic features (AUC = 0.908, 95% CI: 0.848–0.968; Figs. 3A, C, Supplementary Fig. 10A, 10C and Supplementary Table 15), with a significant difference confirmed by DeLong's test ($P < 0.001$; Supplementary Table 8). The RCSA model also showed significant improvements in reclassification (NRI = 0.834, 95% CI: 0.349–1.256, $P < 0.001$) and discrimination (IDI = 0.407, 95% CI: 0.274–0.531, $P < 0.001$), with enhanced calibration (Brier score = 0.053; Fig. 3E). Similar results were observed in Cohort II (Figs. 3B, D, F; Supplementary Fig 10B, 10D; and Supplementary Table 15).

In both cohorts, concentric circle plots indicated that the RCSA model identified a higher proportion of high-risk MLM patients than the clinical model (Cohort I: 11.9% vs. 6.2%; Cohort II: 13.1% vs. 11.3%, Fig. 3G). Decision curve and clinical impact analyses further confirmed the superior clinical utility of the RCSA model in both cohorts (Fig. 3H-I; Supplementary Fig. 10G–H). Survival analysis revealed significantly better 5-year OS for low-risk patients compared to high-risk patients as stratified by the RCSA model (Cohort I: 57.6% vs. 41.9%, $P = 0.009$, Supplementary Fig. 10E; Cohort II: 47.5% vs. 29.6%, $P = 0.012$, Supplementary Fig. 10F). The RCSA model achieved robust discriminative performance for survival prediction across the internal validation cohorts, with concordance indices of 0.633 (95% CI: 0.573–0.692) in cohort I and 0.650 (95% CI: 0.596–0.704) in cohort II (Supplementary Table 11).

A sensitivity analysis combining both internal validation cohorts and an independent dataset further confirmed the model's robustness: the RCSA model identified more high-risk MLM patients than the clinical model (51.2% vs. 28.0%), representing a 23.2% increase in detection, with a 5.7% reduction in low-risk misclassification (Fig. 3J).

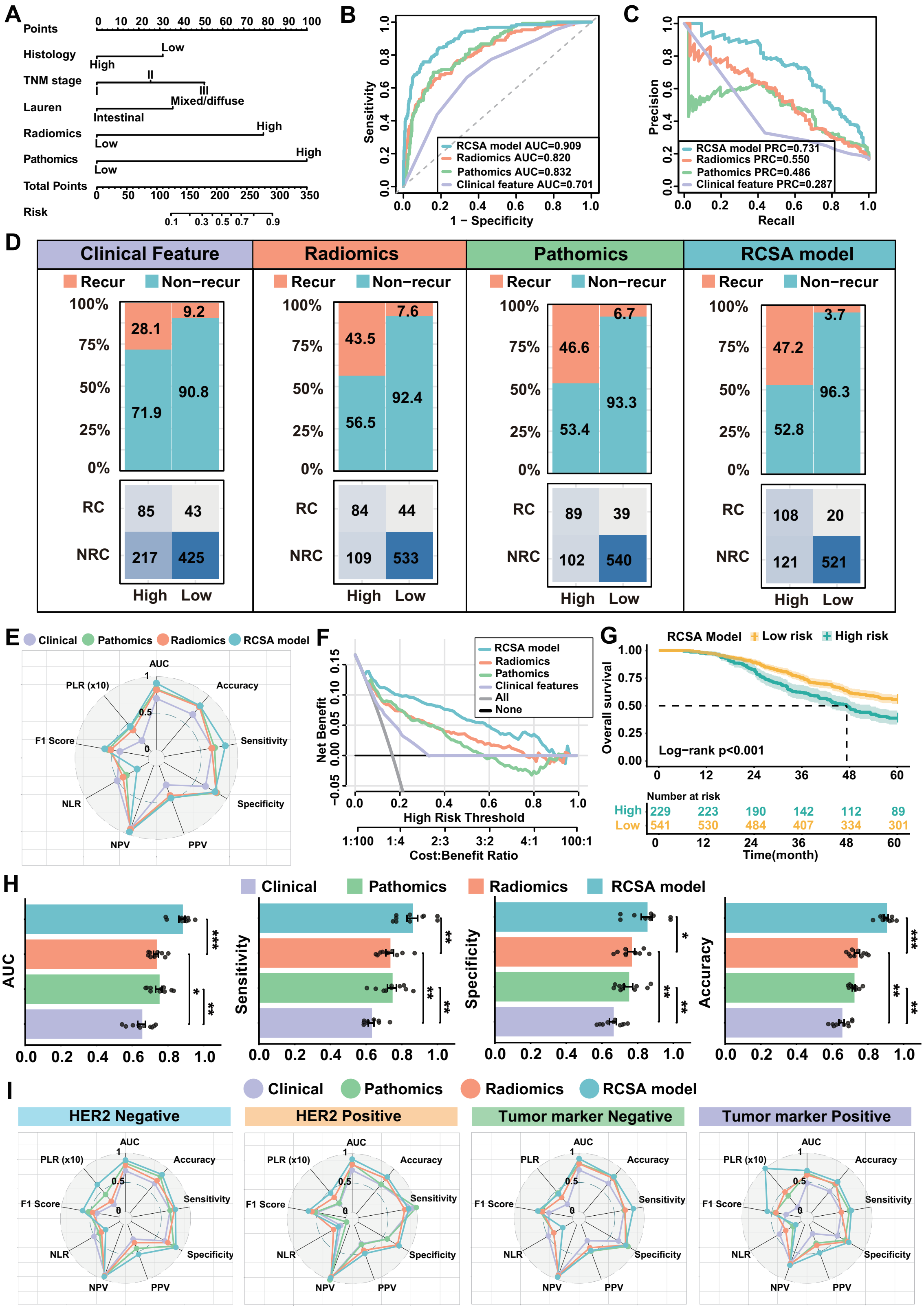
External validation and clinical applicability of the RCSA model

To further assess the clinical applicability and broad generalizability of the RCSA model, we included three independent external validation cohorts. The first cohort (External Validation I) consisted of 257 LAGC patients from three medical centers in northern China (SJZPH, BDCH, HSPH), while the second (External Validation II) included 191 patients from two centers in southern China (WHRH, NJJLH), primarily to account for geographical and regional differences within the Chinese population. The clinicopathological characteristics of these external cohorts are summarized in Supplementary Table 16. Furthermore, to evaluate the robustness of the model across publicly available data sources, we incorporated an independent external validation cohort comprising 41 LAGC patients retrieved from TCIA. These patients were enrolled following the same stringent inclusion and exclusion criteria applied to the multi-center clinical cohorts.

In External Validation I, the RCSA model outperformed models based on individual features in predicting postoperative MLM, with an AUC of 0.890 (95% CI: 0.828–0.951; Fig. 4A, B; Supplementary Fig. 11B; and Supplementary Table 17). Compared with the clinical model, the RCSA model showed significant improvement in NRI (0.882, 95% CI: 0.568–1.212, $P < 0.001$) and IDI (0.364, 95% CI: 0.277–0.446, $P < 0.001$; Supplementary Table 8). Calibration analysis further confirmed its predictive accuracy (Brier score = 0.071; Fig. 4C). Similar results were observed in External Validation II (Fig. 4D–F; Supplementary Fig. 11B; Supplementary Table 17). Decision curve analysis demonstrated the highest net clinical benefit for the RCSA model in both external cohorts (Supplementary Fig. 11C, D), which was further supported by clinical impact curve analysis (Supplementary Fig. 11E, F). Concentric circle plots revealed that the RCSA model identified a greater proportion of high-risk MLM patients compared to the clinical model in both cohorts (External I: 14.8% vs. 10.1%, Fig. 4I; External II: 14.2% vs. 7.3%, Fig. 4J).

In the TCIA validation cohort, the RCSA model demonstrated robust predictive performance with an AUC of 0.862 (95%CI: 0.721-1.000) (Supplementary Fig. 12A–C; Supplementary Table 17). Compared to the clinical model, RCSA achieved significantly improved accuracy (DeLong' test $P = 0.004$) and superior risk stratification, evidenced by an NRI of 0.695 and an IDI of 0.382 (Supplementary Table 8). Performance across models was further quantified through confusion matrices and radar charts, where RCSA maintained a high sensitivity of 0.833 and an NPV of 0.964 (Supplementary Fig. 12D and Supplementary Table 17). Additionally, concentric donut plots confirmed that the RCSA model more effectively identified high-risk MLM patients (12.2%) compared to the clinical model (Supplementary Fig. 12E).

Follow-up survival data showed significantly worse 5-year OS among high-risk patients compared to low-risk patients, as stratified by the RCSA model (External I: 33.8% vs. 51.9%, $P = 0.002$, Fig. 4K; External II: 37.9% vs. 54.1%, $P = 0.012$, Fig. 4L and Supplementary Fig. 12H). RCSA



achieved robust discriminative performance across all external validation cohorts, with concordance indices of 0.618 (95% CI: 0.572–0.664) in External validation I, 0.646 (95% CI: 0.592–0.699) in External validation II (Supplementary Table 11). A sensitivity analysis combining the two external validation cohorts into a single independent dataset showed that the RCSA model identified 21.5% more high-risk MLM patients than the clinical model, while reducing the proportion of misclassified low-risk patients by 8.0%, potentially minimizing unnecessary follow-up imaging (Supplementary Fig. 11G).

Prospective validation of the RCSA Model

To evaluate the RCSA model's performance in a prospective setting, we retrospectively analyzed data from a multicenter prospective clinical study (NCT 02555358). A total of 257 LAGC patients were

Fig. 2 | Construction and performance evaluation of the multimodal RCSA model in the training cohort. **A** A multimodal nomogram (RCSA model) developed by integrating independent clinical predictors (histology, pTNM stage, and Lauren classification) with radiomic and pathomic signatures for predicting MLM in patients with LAGC. **B** ROC curves comparing the discriminative performance of the RCSA model (AUC = 0.909) against unimodal and clinical-only models. **C** Precision-recall curves demonstrating the model's reliability in handling imbalanced event distribution. **D** Radar chart illustrating multiple performance metrics (including Accuracy, Sensitivity, Specificity, PPV, NPV, PLR, NLR and F1 score) across different models. **E** Confusion matrices showing the classification results and detection rates for the RCSA-defined high- and low-risk groups. RC, cases that developed metachronous liver metastasis; NRC, cases that did not. **F** Decision curve analysis evaluating the clinical net benefit of the RCSA model across various threshold probabilities. **G** Kaplan–Meier curves showing overall survival according to the low- and high-risk groups defined by the RCSA model using the optimal Youden index cutoff. Solid lines indicate the estimated survival probabilities, and shaded regions represent 95% confidence intervals. The low-risk and high-risk groups included 541 and 229 patients, respectively. Survival distributions were compared using a two-sided log-rank test (exact $P = 4.413 \times 10^{-6}$). **H** Tenfold cross-validation was used to assess the stability of model performance. Each dot corresponds to the result obtained from one validation fold, with the 770 independent patients partitioned into ten mutually exclusive subsets. Bars show the average value across folds, and error bars denote the standard deviation (SD). Differences between the RCSA model and the Pathomics, Radiomics and Clinical feature models were evaluated using two-sided Wilcoxon rank-sum tests. The exact P values for RCSA versus Pathomics, RCSA versus Radiomics and RCSA versus Clinical feature were, respectively: AUC, $P = 5.80 \times 10^{-4}$, $P = 3.30 \times 10^{-4}$ and $P = 1.83 \times 10^{-4}$; Sensitivity, $P = 0.01696$, $P = 0.00715$ and $P = 1.76 \times 10^{-4}$; Specificity, $P = 0.03108$, $P = 0.03742$ and $P = 9.99 \times 10^{-4}$; and Accuracy, $P = 1.82 \times 10^{-4}$, $P = 1.82 \times 10^{-4}$ and $P = 1.81 \times 10^{-4}$. Source data are provided as a Source Data file. **I** Radar charts showcasing the robust predictive performance of the RCSA model in specific subgroups stratified by HER2 status and serum tumor markers. Data are presented as mean ± SD for cross-validation where applicable; $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. AUC, area under the receiver operating characteristic curve; F1 score harmonic mean of precision and recall; HER2 human epidermal growth factor receptor 2; NLR negative likelihood ratio; NPV negative predictive value; PLR positive likelihood ratio; PPV positive predictive value; PRC precision–recall curve; RCSA radiopathomics-clinical stratification assessment. Source data are provided as a Source Data file.

included based on strict inclusion and exclusion criteria: 93 underwent primary surgery, 86 received neoadjuvant XELOX chemotherapy (oxaliplatin + capecitabine), and 78 received the DOX regimen (docetaxel + oxaliplatin + capecitabine). Clinical characteristics are summarized in Supplementary Table 18, with clinical outcomes previously detailed in our published work[27].

In the surgery cohort, the RCSA model outperformed clinical, radiomic, and pathomic models in predicting postoperative MLM, with an AUC of 0.898 (95% CI: 0.816–0.980; Figs. 5A, B, D; Supplementary Fig. 13A, B and Supplementary Table 19). Compared with the clinical model, the RCSA model achieved significant improvements in NRI (0.630) and IDI (0.284), with similar advantages observed when compared to radiomic and pathomic models (Supplementary Table 8). Concentric circle plots showed that the RCSA model identified more high-risk MLM patients than the clinical model (10.8% vs. 8.6%; Fig. 5F). Decision curve analysis and clinical impact curves further confirmed its superior clinical utility in the primary surgery cohort (Fig. 5C; and Supplementary Fig. 13C). Further survival analysis found that the 5-year OS of patients in the low-risk group of the RCSA model was significantly better than that of patients in the high-risk group (57.7% vs. 31.8%; $P = 0.009$; Fig. 5K). In the prospective direct surgery cohort, RCSA achieved a concordance index of 0.632 (95% CI: 0.553–0.711) for survival prediction (Supplementary Table 11).

A similar analysis in the neoadjuvant chemotherapy cohorts demonstrated consistently strong predictive performance of the RCSA model (AUC = 0.786, 95% CI: 0.680–0.893; Figs. 5E, G–J; Supplementary Fig. 13D, 13F, G; and Supplementary Table 8, 19), with predictions correlating significantly with treatment response (Supplementary Fig. 13E). In the prospective neoadjuvant chemotherapy cohort, RCSA stratified patients into groups with different 5-year OS (low-risk 55.1% versus high-risk 43.9%, $P = 0.044$; Fig. 5L), with a concordance index of 0.602 (95% CI: 0.539–0.666) (Supplementary Table 11).

Model explainability and individualized prediction validation based on SHAP analysis

We applied SHapley Additive exPlanations (SHAP) to examine feature contributions in the radiomics and pathomics models. As shown in the SHAP Bee Swarm Plots (Fig. 6A–L and Supplementary Fig. 14), the detailed impact patterns of key features on the model's prediction results are revealed: high feature values (warm-colored dots) for both Radiomics (e.g., Feature-268, Feature-471) and Pathomics (e.g., Feature-437, Feature-232) features primarily aggregate in the positive SHAP value region, indicating a significant contribution to high-risk prediction; conversely, low feature values (cool-colored dots) tend toward a negative contribution. Concurrently, the SHAP Mean Bar Plots (Supplementary Fig. 15) identify the core driving features: Feature-268, Feature-471, and Feature-108 consistently hold the highest mean absolute SHAP values in the Radiomics model, while Feature-437, Feature-232, and Feature-188 dominate the Pathomics model. Similar patterns of feature importance were observed across cohorts.

To provide clinical interpretability for the model, we present two patient cases with different outcomes for individualized prediction explanation in Fig. 6M. This visualization integrates the patient's CT images, H&E sections, and corresponding feature maps, utilizing SHAP values to quantify the contribution of each feature to the final prediction. In the case of Patient 1 (46-month recurrence-free follow-up), the feature contribution plot leans toward a negative prediction ($f(x) = -0.174$); for example, the radiomic feature Feature-268 provides the dominant negative contribution (SHAP = −0.124), supporting the low recurrence risk assessment. Conversely, in the case of Patient 2 (21-month liver recurrence), the plot shifts toward a positive prediction ($f(x) = +0.007$), with Feature-268 providing the dominant positive contribution (SHAP = +0.051), clearly supporting the high recurrence risk.

A cost–benefit analysis based on current hospital follow-up protocols (Supplementary Fig. 16) showed improved resource utilization compared with the clinical model. In clinical terms, RCSA increased the identification rate of MLM by 9.2% (92.3% versus 83.1%) and reduced missed MLM cases (false negatives) by 24 patients (from 44 to 20).

Molecular and immune landscape differences between risk groups

To investigate the molecular differences between high- and low-risk groups, we conducted transcriptome sequencing and identified distinct gene expression profiles (Fig. 7A). Gene Set Variation Analysis (GSVA) and Gene Set Enrichment Analysis (GSEA) consistently revealed a significant enrichment of core immune-related pathways, including interferon-γ and interferon-α responses, alongside inflammation-related pathways (IL6/JAK/STAT3, TNF-α, IL2/STAT5 signaling, and general inflammatory response), predominantly in the low-risk cohort (Fig. 7B, C and Supplementary Fig. 17A).

We next examined the immune microenvironment. ESTIMATE analysis confirmed that the low-risk group exhibited significantly higher Stromal, Immune, and overall ESTIMATE scores (Supplementary Fig. 17B), indicative of a more active and "hot" tumor microenvironment. CIBERSORT cell infiltration analysis further validated this immune activity, showing that the abundance of T cells (especially CD8 + T cells) was significantly higher in the low-risk group (Fig. 7D, F), which is consistent with the results of MCPcounter (Supplementary Fig. 17C, 17D). Furthermore, expression profiling of immune

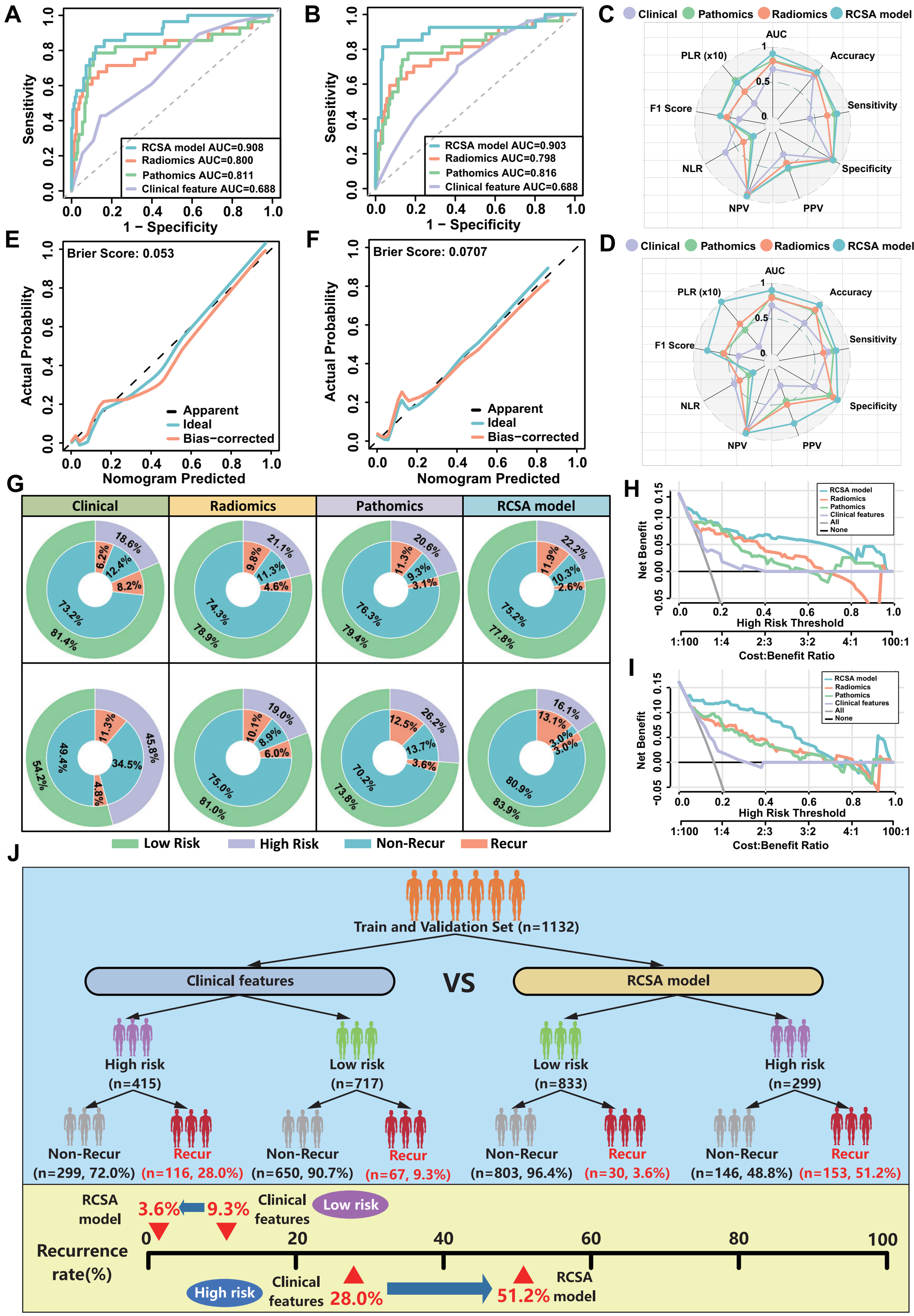
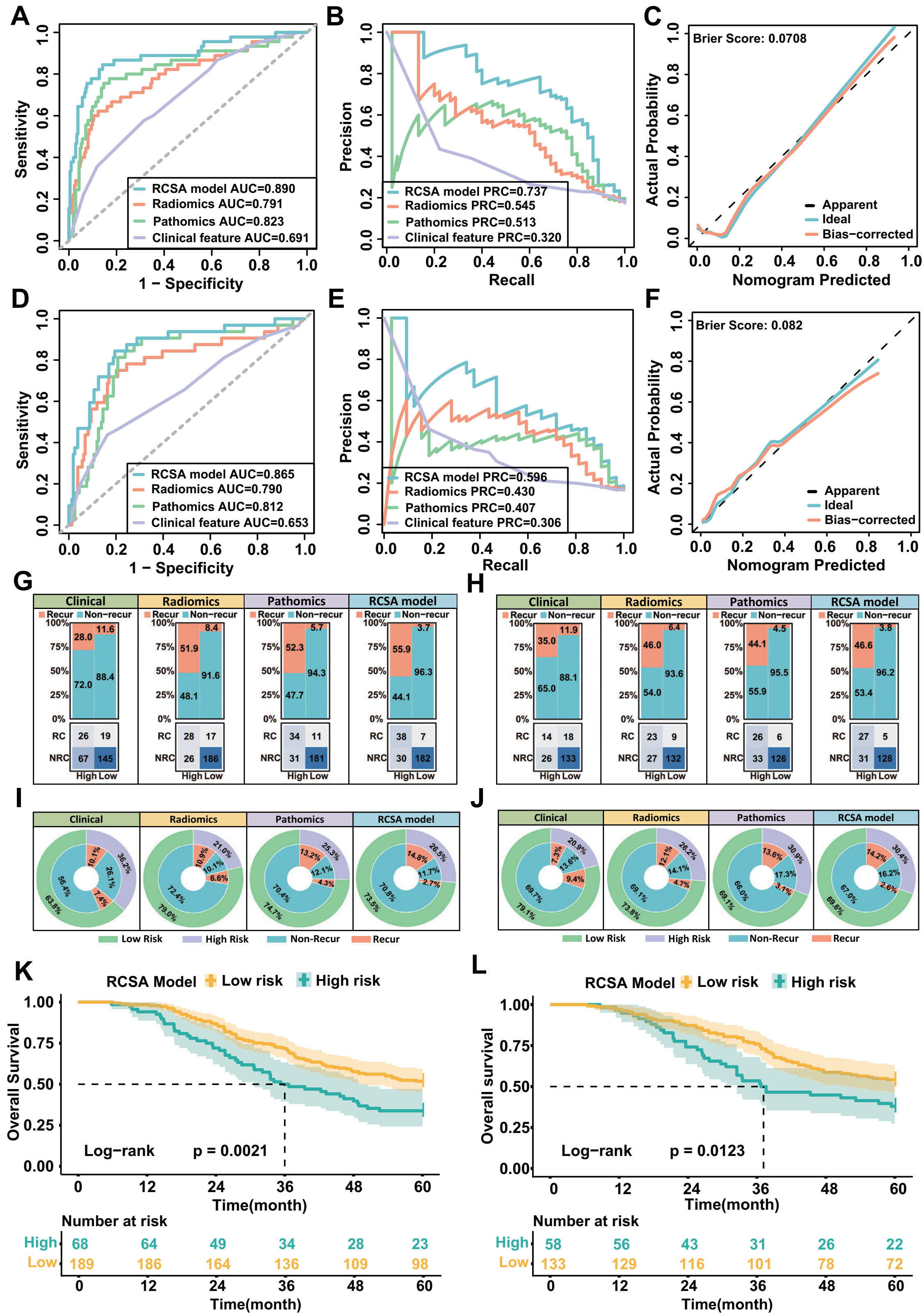


Fig. 3 | Internal validation and clinical utility of the RCSA model in two independent cohorts. A, B ROC curves of the RCSA model compared with clinical, radiomic, and pathomic models in Internal Validation Set I (**A**) and Set II (**B**). **C, D** Radar charts summarizing the multi-dimensional performance metrics for all models across the two validation cohorts. **E, F** Calibration curves showing the high agreement between RCSA-predicted probabilities and actual MLM incidence (Brier score = 0.053 for Set I; 0.071 for Set II). **G** Concentric circle plots visualizing the clinical benefit and risk stratification, demonstrating that the RCSA model identifies a higher proportion of true-positive MLM cases compared to the clinical-only model. **H, I** Decision curve analysis illustrating that the RCSA model provides the highest net clinical benefit across a wide range of threshold probabilities in both cohorts. **J** A comprehensive flow diagram illustrating the reclassification improvement of the RCSA model over clinical features, highlighting a significant increase in high-risk detection (from 28.0% to 51.2%) and a reduction in low-risk misclassification. Source data are provided as a Source Data file. Schematic illustrations in this figure were created with BioRender.com.



checkpoint genes showed significantly elevated expression in the low-risk group, particularly genes associated with T cell co-stimulatory activation (Fig. 7G and Supplementary Fig. 17E).

Extending this biological correlation, a multicellular ecosystem analysis demonstrated that communities CE3, CE9, and CE10 were more abundant in the low-risk group (Fig. 7E and Supplementary Fig. 18), and scoring based on IFN-γ and effector T cell-related gene signatures confirmed significantly elevated expression of IFNγ-6, IFNγ-18, effector T cells, and IFNγ/T cell signaling signatures in low-risk patients (Fig. 7H). Collectively, these robust immunological findings provide a strong molecular basis for the RCSA model's prognostic capability and suggest that low-risk patients, characterized by a potent anti-tumor immune response, may be more responsive to adjuvant immunotherapy.

Fig. 4 | Multi-center external validation of the RCSA model across diverse geographical cohorts. **A** ROC curves in External Validation Set I. **B** Precision-recall curves in External Validation Set I. **C** Calibration curve for External Validation Set I (Brier score = 0.071). **D** ROC curves in External Validation Set II. **E** Precision-recall curves in External Validation Set II. **F** Calibration curve for External Validation Set II (Brier score = 0.082). **G** Confusion matrices for Set I. **H** Confusion matrices for Set II. **I** Concentric circle plots for Set I. **J** Concentric circle plots for Set II. **K** Kaplan–Meier curves for overall survival in External Validation Set I, stratified by the RCSA-defined low- and high-risk groups using the cutoff derived from the training cohort. Solid lines indicate the estimated survival probabilities, and shaded bands represent the 95% confidence intervals. The low-risk and high-risk groups included $n = 189$ and $n = 68$ independent patients, respectively. Survival differences were evaluated using a two-sided log-rank test (exact $P = 0.002072$). **L** Kaplan–Meier curves for overall survival in External Validation Set II, stratified by the same RCSA cutoff. Solid lines indicate the estimated survival probabilities, and shaded bands represent the 95% confidence intervals. The low-risk and high-risk groups included $n = 133$ and $n = 58$ independent patients, respectively. Survival differences were evaluated using a two-sided log-rank test (exact $P = 0.012295$). AUC area under the receiver operating characteristic curve; PRC precision–recall curve; RCSA radiopathomics-clinical stratification assessment. Source data are provided as a Source Data file.

Discussion

In this study, we developed and validated a comprehensive RCSA model for predicting MLM in patients with LAGC. The RCSA model integrates clinical parameters, radiomic features derived from preoperative CT scans, and deep learning-based pathomic features from histopathological slides. The model demonstrated strong predictive performance across multiple cohorts, including internal, external, and prospective validation datasets, significantly outperforming models that rely solely on clinical, radiomic, or pathological features. Notably, the RCSA model achieved consistent robustness and accuracy across distinct geographical regions and treatment strategies, reinforcing its potential clinical applicability. Furthermore, transcriptomic analysis revealed substantial differences in molecular pathways and immune infiltration between high-risk and low-risk patient groups, highlighting the biological relevance of the RCSA-based stratification and suggesting potential therapeutic implications, particularly concerning immunotherapy responsiveness.

The predictive capacity of the RCSA model exceeds previously reported clinical and unimodal prediction models. Prior studies primarily used either clinical or radiomic factors separately to predict gastric cancer prognosis. For example, Cao et al. constructed a radiomics model based on preoperative CT images, which accurately predicted recurrence and survival in gastric cancer patients[28]. Their model achieved a respectable predictive value (AUC ~0.80–0.85), but its generalizability remained uncertain due to lack of external validation across diverse cohorts. In contrast, the RCSA model presented herein demonstrated significantly improved predictive power (AUC > 0.89 consistently), validated through multiple external cohorts across different regional contexts. This enhanced generalizability is likely attributed to the integration of pathological features, adding crucial information that radiomics or clinical data alone might overlook, such as microscopic tumor heterogeneity and TME complexity. Chen et al. also developed pathology-based predictive models utilizing machine learning approaches, yet they encountered notable accuracy reductions upon external validation[29]. This phenomenon underscores the importance of multimodal integration, as observed in our RCSA model, which leverages complementary information across radiomics and pathomics, thereby maintaining high predictive accuracy even in external validation scenarios.

The RCSA model's robustness was also demonstrated by its effectiveness in patients undergoing neoadjuvant chemotherapy, a clinically relevant subgroup. Cui et al. previously reported that neoadjuvant chemotherapy significantly alters radiomic characteristics, potentially diminishing the predictive reliability of preoperative imaging-based models[30]. Interestingly, our RCSA model maintained robust performance in the cohort receiving neoadjuvant chemotherapy (AUC = 0.786). This suggests that the integration of deep learning-based pathological features effectively captures chemotherapy-induced histopathological changes, compensating for limitations of radiomic features alone. Additionally, geographical disparities in gastric cancer biology and management have been acknowledged in the literature, notably by Lu et al., who indicated significant regional variability in metastasis incidence and tumor aggressiveness[31]. By incorporating data from northern and southern China, our model validated its broad applicability across distinct epidemiological settings, thereby affirming the generalizability of RCSA across diverse clinical populations. The improved predictive performance of RCSA may translate into meaningful clinical and economic benefits. The observed reduction in false-negative cases could help reduce delays in the detection of postoperative MLM, thereby supporting timely clinical intervention. Conversely, a lower false-positive rate may reduce the need for intensive follow-up in patients at low risk, with potential implications for decreasing unnecessary procedures and optimizing resource utilization.

The exploration of molecular mechanisms underlying the predictive stratification provided crucial insights into the biological basis of RCSA-derived risk groups. RNA sequencing analysis demonstrated distinct transcriptional profiles between low-risk and high-risk groups, particularly highlighting significant enrichment of immune-related pathways in the low-risk group. Pathways such as interferon-γ, interferon-α responses, and various inflammation-associated signaling cascades (IL6/JAK/STAT3, IL2/STAT5) were prominently activated among patients classified as low risk. These results are consistent with recent studies suggesting immune activation as a favorable prognostic indicator and potential biomarker for immunotherapy responsiveness in gastric cancer[32–34]. Furthermore, detailed immune microenvironment analysis using ESTIMATE, CIBERSORT, and MCPcounter indicated robust infiltration of $CD8^+$ T cells in low-risk tumors. This finding aligns with the prevailing understanding that enhanced cytotoxic T-cell infiltration correlates positively with prognosis and may predict improved response to immunotherapy. Elevated expression of immune checkpoint genes, notably those involved in T-cell co-stimulation, further supports the hypothesis that low-risk patients identified by the RCSA model may preferentially benefit from immunotherapeutic strategies. Based on these transcriptomic findings, immune-inflamed, low-risk tumors identified by RCSA may be associated with MSI-H status or higher TMB, both of which have been linked to enhanced antitumor immune responses and favorable outcomes in gastric cancer[35]. Conversely, immune-cold, high-risk tumors may exhibit a higher prevalence of alterations in DDR pathways, which could contribute to genomic instability and metastatic potential[36]. Future integration of genomic features with radiopathomic profiles may facilitate a more comprehensive radiopathogenomics framework for MLM prediction. Beyond prognosis, the RCSA model may support a risk-adapted surveillance framework. Patients classified as high risk could undergo more frequent imaging assessments (for example, CT or MRI every 3–4 months), whereas patients at low risk may be suitable for standard surveillance intervals.

Despite these notable strengths, several limitations of this study should be recognized. First, the retrospective nature of the primary dataset poses potential selection biases, though mitigated by prospective validation and rigorous multicenter analyses. Additionally, despite extensive validation, further large-scale, prospective trials across international centers are warranted to solidify the model's global applicability. Second, the complexity inherent in multimodal data integration, especially deep learning-based pathological analysis, poses significant challenges for practical clinical implementation. Developing streamlined, user-friendly computational tools and

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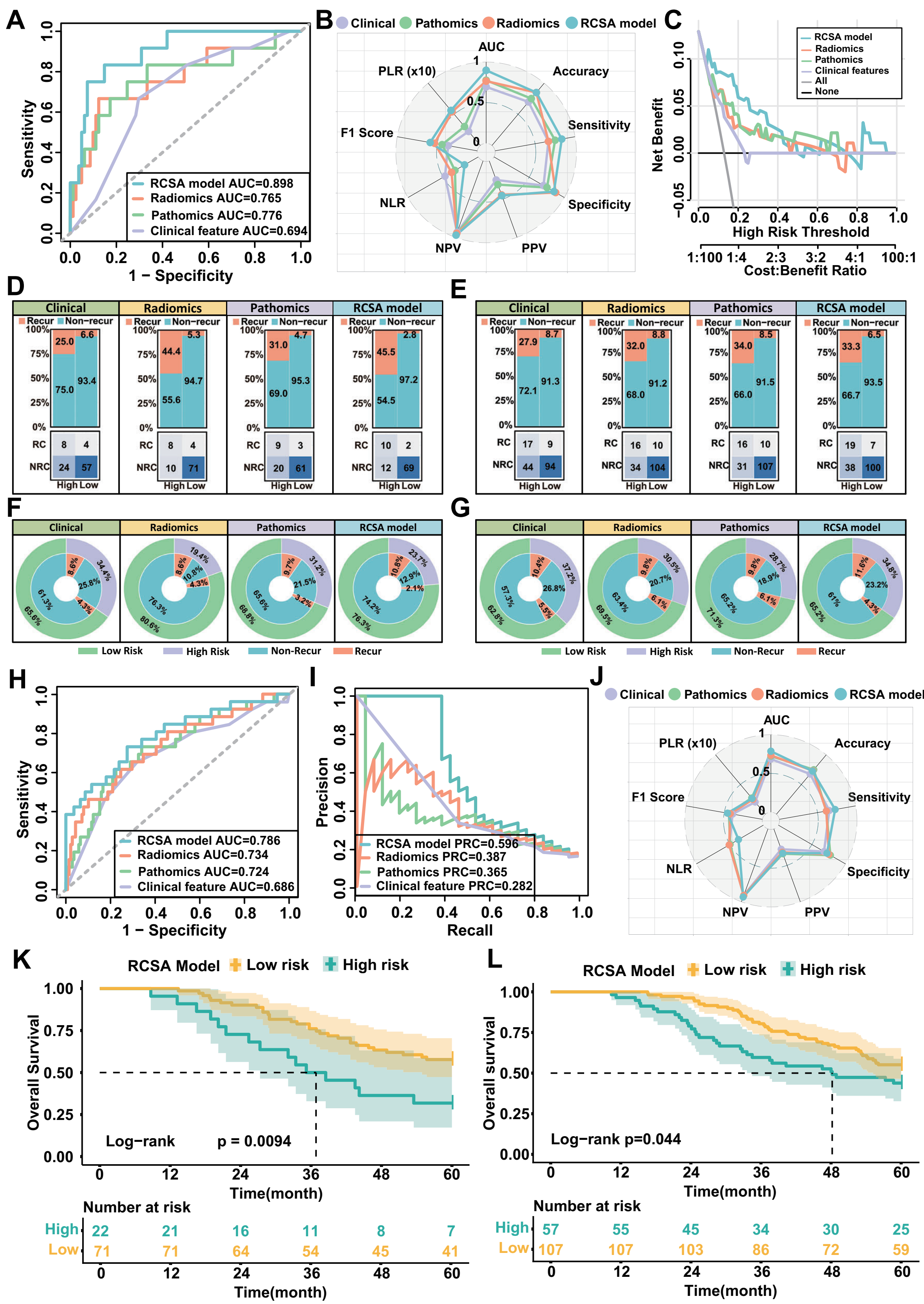


automated workflows will be essential for real-world adoption of the RCSA model. To address this, we propose a multi-layered integration strategy centered on developing a browser-based interface compatible with existing hospital picture archiving and communication systems and laboratory information systems, aiming to enable automated data acquisition, standardized feature extraction, and seamless reporting to the EHR system. Finally, although comprehensive, the current model lacks genomic characterization. Future studies integrating genomic profiling could further enhance predictive accuracy and personalize therapeutic strategies by providing deeper insights into underlying genetic determinants of MLM risk.

Fig. 5 | Prospective validation of the RCSA model in a multicenter clinical study cohort. **A** ROC curves in the primary surgery group, showing the superior discriminative power of the RCSA model (AUC = 0.898). **B** Radar chart summarizing performance metrics (Accuracy, Sensitivity, Specificity, etc.) for the primary surgery group. **C** Decision curve analysis demonstrating the highest clinical net benefit for the RCSA model in the surgery cohort. **D** Confusion matrices and classification results for the primary surgery group. **E** Confusion matrices for the neoadjuvant chemotherapy (NACT) group. **F** Concentric circle plots visualizing risk stratification and clinical benefit in the primary surgery group. **G** Concentric circle plots for the NACT group. **H** ROC curves in the NACT group, highlighting the robust performance of the RCSA model (AUC = 0.786) even after chemotherapy. **I** Precision-recall curves in the NACT group, demonstrating superior predictive stability. **J** Radar chart summarizing multi-dimensional performance metrics for the NACT group. **K** Kaplan–Meier curves for overall survival in the primary surgery group, stratified according to the RCSA-defined low- and high-risk groups using the cutoff derived from the training cohort. Solid lines indicate the estimated survival probabilities and shaded bands represent the 95% confidence intervals. The low-risk and high-risk groups included $n = 71$ and $n = 22$ independent patients, respectively. Survival distributions were compared using a two-sided log-rank test (exact $P = 0.009414$). **L** Kaplan–Meier curves for overall survival in the NACT group, stratified using the same RCSA cutoff. Solid lines indicate the estimated survival probabilities and shaded bands represent the 95% confidence intervals. The low-risk and high-risk groups included $n = 107$ and $n = 57$ independent patients, respectively. Survival distributions were compared using a two-sided log-rank test (exact $P = 0.044057$). AUC area under the receiver operating characteristic curve; NACT, neoadjuvant chemotherapy; PPV positive predictive value; NPV negative predictive value; PLR positive likelihood ratio; NLR negative likelihood ratio; RCSA radiopathomics-clinical stratification assessment. Source data are provided as a Source Data file.

In conclusion, our study established and validated a multimodal RCSA prediction model that improves the accuracy of postoperative MLM risk assessment in patients with LAGC. The integration of clinical, radiomic, and pathomic data not only enhanced predictive capabilities but also provided insights into tumor biology, particularly concerning immune modulation. This model demonstrated superior performance across geographically diverse cohorts and different treatment regimens, underscoring its broad clinical utility. While prospective validation, further technological refinement and the integration of genomic data remain necessary, the model is intended to serve as a postoperative risk stratification tool. This approach may inform personalized adjuvant therapy planning and surveillance strategies, with the potential to improve outcomes in patients with LAGC.

Methods

Ethics approval and consent to participate

This study was conducted in accordance with the Declaration of Helsinki and all applicable ethical regulations. This study was approved by the Institutional Review Board of the Fourth Hospital of Hebei Medical University (approval no. 2025KT150). The requirement for written informed consent was waived because the study involved the retrospective analysis of previously collected clinical, imaging and pathological data that were anonymized before analysis, involved no additional patient contact or intervention, and posed no more than minimal risk to participants. Participants contributing tumor tissue to the prospective transcriptomic cohort each provided written informed consent before sample collection. Patients in the retrospective cohorts received no compensation, and those in the prospective sequencing cohort received no financial compensation beyond their routine clinical care. Participant sex was obtained from the medical records as self-reported; both male and female patients with locally advanced gastric cancer were included, and no analyses were restricted to a single sex or gender. This prediction-model study is reported in accordance with the TRIPOD + AI reporting guideline[37], and the underlying multicenter observational cohort is additionally reported following the STROBE statement[38].

Study cohort

We retrospectively analyzed data from patients with LAGC treated between January 2012 and December 2019 at four medical centers in Hebei Province: the Fourth Hospital of Hebei Medical University (FHHMU), Shijiazhuang People's Hospital (SJZPH), Baoding Central Hospital (BDCH), and Hengshui People's Hospital (HSPH). Data from two additional centers in southern China, Renmin Hospital of Wuhan University (WHRH) and Nanjing Jinling Hospital (NJJLH), were also included for external validation (Fig. 1).

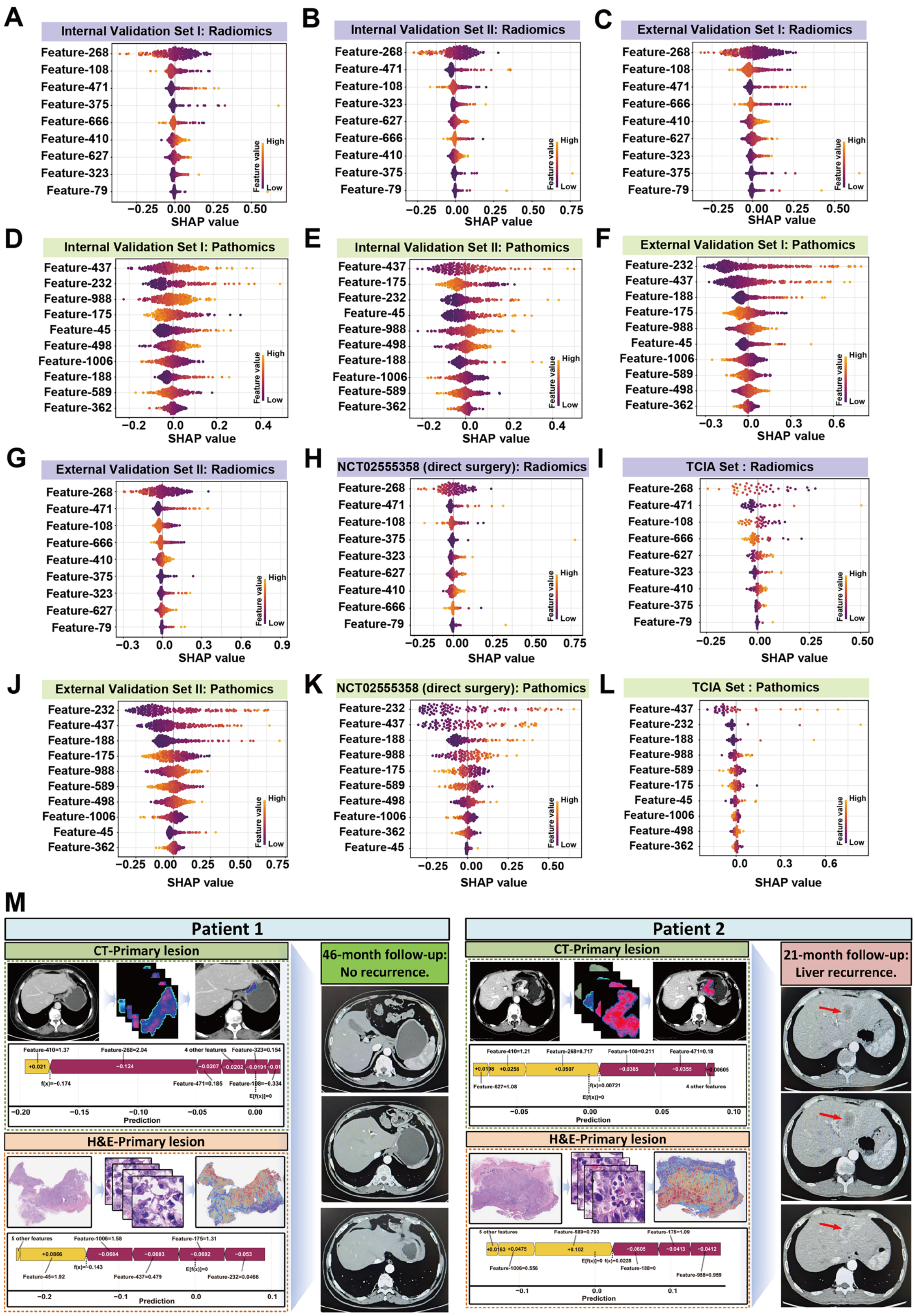
A total of 1878 LAGC patients from six centers and The Cancer Imaging Archive (TCIA) public database were enrolled according to strict inclusion and exclusion criteria (**Supplementary Methods**). The cohort was divided into three datasets: a training cohort of 770 patients from the FHHMU (Center 1, 2014–2017); an internal validation cohort of 362 patients, further split by diagnosis time into Internal Validation I (2012–2013, $N = 194$) and Internal Validation II (2018–2019, $N = 168$); and an external validation cohort of 489 patients from five centers and a public dataset. The external cohort was subdivided based on geographic region: External Validation I (Northern China: Centers 2–4, $N = 257$) and External Validation II (Southern China: Centers 5–6, $N = 191$). To further evaluate the model's generalizability across diverse clinical settings, an additional independent validation was conducted using a public dataset from The Cancer Imaging Archive (TCIA). By applying the same inclusion and exclusion criteria as the primary cohorts, a total of 41 LAGC patients were enrolled from the TCIA database for analysis.

To assess clinical applicability, we retrospectively analyzed a separately selected cohort from a registered multicenter randomized controlled trial (ClinicalTrials.gov, NCT02555358; first posted 21 September 2015; https://clinicaltrials.gov/study/NCT02555358), independent of the retrospective training and validation datasets. The trial compared neoadjuvant chemotherapy regimens (docetaxel, oxaliplatin plus capecitabine [DOX] versus oxaliplatin plus capecitabine [XELOX]) with surgery alone, using pathological complete response as the prespecified primary endpoint. The present evaluation of metachronous liver metastasis prediction represents a post-hoc, exploratory secondary use of this trial cohort and was not part of the registered protocol or analysis plan; no hypothesis regarding the RCSA model was prespecified for this population, and it is reported here solely to assess the generalizability of the RCSA model in an independently enrolled cohort. This cohort included 257 LAGC patients, of whom 93 underwent primary surgery and 164 received neoadjuvant chemotherapy (NACT). Although the training and retrospective cohorts strictly excluded NACT patients, all patients in this cohort were screened using consistent inclusion/exclusion criteria appropriate for this study population. Inclusion of the NACT subgroup enabled evaluation of model performance across different initial treatment strategies.

To explore the biological relevance of the model, we prospectively collected fresh tumor tissues from 82 LAGC patients for transcriptomic sequencing and functional enrichment analysis. These samples were fully consumed and destroyed during sequencing, with none retained for further analysis or sharing. The protocols for RNA processing and sequencing are provided in detail in the **Supplementary Methods**.

Clinical outcomes

Postoperative management, including adjuvant chemotherapy and systematic follow-up, was performed for all patients in accordance with the recommendations of the Chinese Society of Clinical Oncology (CSCO)[39] and the National Comprehensive Cancer Network (NCCN)[40]



guidelines. MLM was defined as metachronous liver metastasis arising more than six months after curative surgery for LAGC, with liver recurrence confirmed by clinical symptoms, reoperation, or imaging (abdominal CT, MRI, ultrasound, or PET/CT). The primary endpoint was postoperative MLM and the secondary endpoint was OS, defined as the interval from surgery to death from any cause.

Image acquisition, segmentation, and radiomics feature selection

All patients underwent contrast-enhanced abdominal CT, and portal venous phase images were retrieved through the Picture Archiving and Communication System. A total of 200 CT scans were used to train and test an automatic 3D tumor segmentation model based on the nnU-Net

Fig. 6 | Model explainability and individualized prediction validation based on SHAP analysis. SHAP bee swarm plots show the contribution of the top features to the model's predictions across six cohorts, presented separately for the radiomic and pathomic modalities, with each radiomic panel and its vertically paired pathomic panel derived from the same cohort. **A**–**C**, **G**–**I** SHAP summary plots for the top radiomic features in Internal Validation Set I (**A**), Internal Validation Set II (**B**), External Validation Set I (**C**), External Validation Set II (**G**), the NCT02555358 direct surgery cohort (**H**), and the TCIA set (**I**). **D**–**F**, **J**–**L** SHAP summary plots for the top pathomic features in the corresponding cohorts, namely Internal Validation Set I (**D**), Internal Validation Set II (**E**), External Validation Set I (**F**), External Validation Set II (**J**), the NCT02555358 direct surgery cohort (**K**), and the TCIA set (**L**). In every plot each point represents an individual patient, its position along the horizontal axis gives the SHAP value and therefore the direction and magnitude of that feature's contribution to the predicted MLM risk, and the point color encodes the feature value, with warm colors denoting high values and cool colors denoting low values. Features are ordered from top to bottom by mean absolute SHAP value. **M** Individualized prediction explanations for two representative patients. Patient 1 (left) represents a low-risk case with no recurrence over a 46-month follow-up; the visualization includes CT and H&E primary lesion images with corresponding SHAP force plots leaning toward a negative (low-risk) prediction. Patient 2 (right) represents a high-risk case with liver recurrence at 21 months; the SHAP force plots for both radiomic and pathomic features strongly lean toward a positive (high-risk) prediction. The red arrows in the follow-up CT images indicate the detected metachronous liver metastasis. Note on features: Radiomic features are mathematically defined according to the IBSI standards, where: Feature-108: LoG-filtered (sigma=1.0) first-order skewness. Feature-268: LoG-filtered (sigma=2.0) GLSZM size-zone non-uniformity normalized. Feature-323: LoG-filtered (sigma=3.0) GLDM dependence non-uniformity. Feature-375: Wavelet-LLH filtered first-order energy. Feature-410: Wavelet-LLH filtered GLCM maximum correlation coefficient. Feature-471: Wavelet-LHL filtered first-order kurtosis. Feature-627: Wavelet-LHH filtered GLRLM run variance. Feature-666: Wavelet-HLL filtered first-order skewness. Feature-79: Original GLSZM large area low gray-level emphasis. Pathomic features (e.g., Feature-437, Feature-232) represent high-dimensional latent representations extracted by the UNI deep-learning foundation model and do not have discrete descriptive names. Source data are provided as a Source Data file.

framework. From the resulting three-dimensional regions of interest (3D-ROIs), 1130 radiomics features were extracted and standardized; full details of CT image acquisition, the automated segmentation pipeline, and feature extraction are provided in the **Supplementary Methods**. To identify the most discriminative variables, univariate analysis was followed by two feature-selection strategies, LASSO regression and Random Forest, which yielded 14 and 10 candidate features, respectively (Supplementary Fig. 1A, B). These feature sets were then combined with nine machine learning algorithms to assess different modeling configurations. Among the 18 configurations evaluated, a stepwise logistic regression model built on the LASSO-selected features was the top-performing classifier (Training: AUC = 0.820, 95%CI: 0.779–0.861; Validation: AUC = 0.800, 95%CI: 0.691–0.909; Supplementary Fig. 1C, D; and Supplementary Table 1, 2).

Pathological feature extraction and selection based on the UNI model

Formalin-fixed, paraffin-embedded (FFPE) hematoxylin and eosin (H&E)-stained slides were prepared for all patients and digitized into whole-slide images (WSI). From each slide, 1024 pathomic features were extracted using the UNI[41] tissue-specific encoder (based on the DINOv2 self-supervised algorithm) and aggregated to the slide level; full details of WSI preprocessing, feature extraction, aggregation, and quality control are provided in the **Supplementary Methods**. Discriminative pathological features were then filtered using univariate screening followed by two selection methods (LASSO and Random Forest), yielding 18 and 10 features, respectively (Supplementary Fig. 2A, B). To determine the optimal predictive framework, the two feature sets were paired with nine machine learning architectures, and the resulting 18 classification models were compared. The final pathology-based signature (Supplementary Table 2) was built using a stepwise logistic regression model derived from the LASSO feature set, which achieved the best performance among all tested models (Training AUC = 0.832, 95%CI: 0.796–0.868; Validation AUC = 0.811, 95%CI: 0.699–0.923; Supplementary Fig. 2C–D and Supplementary Table 3).

Development and evaluation of a multimodal model for predicting MLM in LAGC patients

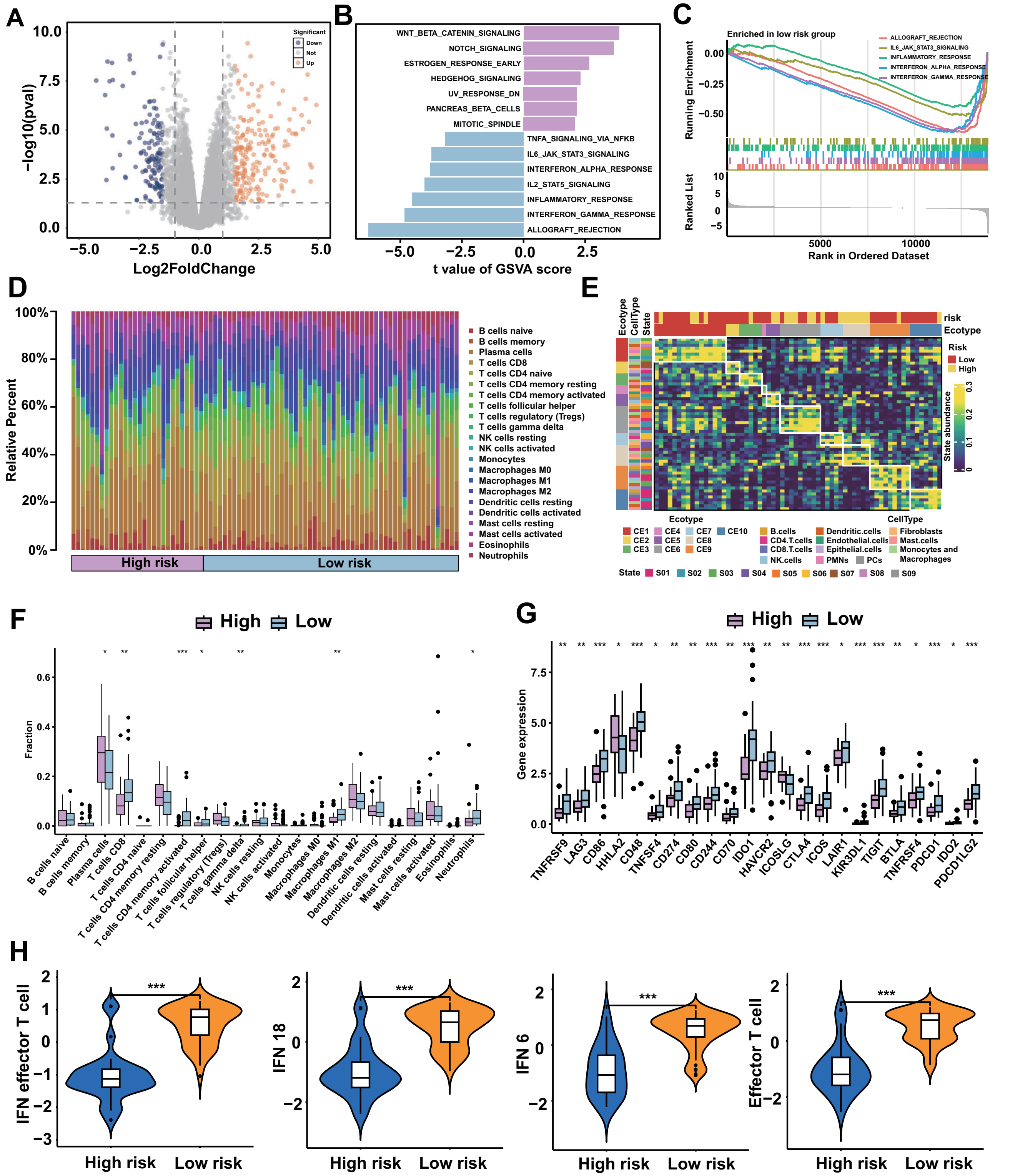
To predict postoperative MLM in patients with LAGC, univariate logistic regression was used to screen clinical variables, and significant variables were entered into a multivariate logistic regression model to identify independent clinical predictors. Model discrimination was assessed using the AUC, accuracy, sensitivity, specificity, and F1 score, and AUCs were compared using the DeLong test; the incremental value of the model was quantified using the net reclassification improvement (NRI) and integrated discrimination improvement (IDI). Calibration was evaluated by plotting calibration curves comparing predicted and observed MLM incidence, and overall performance was summarized using prediction error curves and the integrated Brier score. To benchmark the RCSA model against contemporary methods, we additionally constructed a Transformer-based multimodal deep learning model for comparison[42], with full details provided in the **Supplementary Methods**.

Biological characteristics and immune infiltration

We collected 82 tissue samples from the study cohort for RNA sequencing and exploratory analyses. Functional enrichment analysis, together with the inference of immune infiltration, cell states, and cancer ecosystems using CIBERSORTx and EcoTyper, was performed as detailed in the **Supplementary Methods**. All samples were processed and sequenced in a single batch to minimize technical batch effects. Multiple testing correction was conducted using the Benjamini-Hochberg procedure, with an adjusted *P* value (FDR) < 0.05 considered statistically significant.

Statistics and reproducibility

This multicenter, retrospective observational study developed and validated the RCSA model for predicting MLM after curative resection in LAGC, using cohorts from six Chinese hospitals, a public dataset (TCIA), and a registered randomized controlled trial (NCT02555358). No statistical method was used to predetermine sample size; all consecutive eligible patients at each center were included. Although no a priori calculation was performed, a post-hoc check using the criteria of Riley et al.[43] confirmed adequacy: for the final RCSA model (the radiomic and pathomic signatures entered as single composite predictors plus three independent clinical predictors), an outcome proportion of 0.166 (128 events among 770 patients) and a conservative Cox-Snell R^2 of 0.15 yielded a minimum required size of 384 (18.3 events per predictor parameter), which the development cohort exceeded; LASSO penalization together with internal, external, public, and prospective validation and calibration and bootstrap optimism analyses further guarded against overfitting. Data were excluded only by the predefined eligibility criteria (neoadjuvant chemotherapy in the retrospective cohorts, surgery-related death, loss to follow-up, image artifacts or missing imaging, and missing essential clinical information), with per-reason counts given in the participant flow diagram; no otherwise eligible data were excluded after analysis. The study was not randomized, as patients were grouped by institution and data source in keeping with the observational design. Investigators were not blinded; model development and evaluation were performed computationally using predefined pipelines, and the MLM outcome was



ascertained from radiological follow-up independently of the model predictions.

Statistical analysis

All statistical analyses were conducted using SPSS version 28.0 (IBM) and R version 4.3.3 (http://www.r-project.org). Continuous variables were analyzed using unpaired two-tailed t tests or Mann-Whitney U tests, while categorical variables were evaluated using the Chi-square test or Fisher's exact test. Univariate and multivariate Cox regression analyses were performed to assess the prognostic value of variables for survival. Missing data patterns across clinical variables were examined and were consistent with a missing-at-random (MAR) assumption. The primary analyses were conducted using complete-case analysis (CCA). In addition, sensitivity analyses were performed using multiple imputation by chained equations (MICE) to account for missing clinical data and to evaluate the potential impact of missingness on the results[44,45]. Detailed descriptions of the statistical algorithms are provided in the

Fig. 7 | Molecular and immune characteristics of the RCSA-defined high- and low-risk groups. A Volcano plot showing differentially expressed genes between the high-risk and low-risk groups. Differential expression was assessed using edgeR generalized linear models with likelihood-ratio tests, followed by Benjamini–Hochberg correction for multiple testing. Genes with an absolute log2 fold change greater than 1.5 and a false-discovery-rate-adjusted *P* value below 0.05 were considered differentially expressed. **B** Gene Set Variation Analysis (GSVA) of the 50 MSigDB hallmark gene sets. Bars represent the moderated t statistics obtained using limma, with positive and negative values indicating pathways enriched in the high-risk and low-risk groups, respectively. *P* values were adjusted across the 50 gene sets using the Benjamini–Hochberg method. **C** Gene Set Enrichment Analysis (GSEA) showing selected hallmark pathways enriched in the low-risk group. Normalized enrichment scores and nominal P values were calculated using 1,.000 permutations, and *P* values were adjusted using the Benjamini–Hochberg method. Vertical tick marks indicate the positions of pathway genes in the ranked gene list. **D** Relative proportions of 22 immune-cell populations estimated using CIBERSORT for each tumour sample. Each column represents one independent patient sample, and the stacked fractions within each column sum to 1. **E** EcoTyper analysis showing the abundance of cell states and their organization into cancer ecotypes across tumour samples. Rows represent cell states, columns represent independent patient samples, and the colour scale indicates estimated state abundance. The annotation bars indicate the dominant ecotype, cell type, cell state and RCSA-defined risk group, as defined in the keys within the panel. **F** Comparison of the estimated fractions of 22 CIBERSORT immune-cell populations between the high-risk group ($n = 28$) and low-risk group ($n = 54$). Each data point represents one independent patient and biological replicate. Box centres indicate medians, box bounds represent the 25th and 75th percentiles, and whiskers extend to the most extreme values within 1.5 times the interquartile range; observations outside the whiskers are plotted individually. Groups were compared using two-sided Mann–Whitney *U* tests without adjustment for multiple comparisons. Exact *P* values for significant comparisons were: plasma cells, $P = 0.024199$; CD8 T cells, $P = 0.001189$; activated memory CD4 T cells, $P = 0.000253$; follicular helper T cells, $P = 0.019046$; γδ T cells, $P = 0.001125$; M1 macrophages, $P = 0.008901$; and neutrophils, $P = 0.019644$. **G** Expression of selected immune-checkpoint and immune-regulatory genes in the high-risk ($n = 28$) and low-risk ($n = 54$) groups, shown from left to right as TNFRSF9, LAG3, CD86, HHLA2, CD48, TNFSF4, CD274, CD80, CD244, CD70, IDO1, HAVCR2, ICOSLG, CTLA4, ICOS, LAIR1, KIR3DL1, TIGIT, BTLA, TNFRSF4, PDCD1, IDO2 and PDCD1LG2. Each data point represents one independent patient and biological replicate. Box centres indicate medians, box bounds represent the 25th and 75th percentiles, and whiskers extend to the most extreme values within 1.5 times the interquartile range; observations outside the whiskers are plotted individually. Groups were compared using two-sided Mann–Whitney *U* tests without adjustment for multiple comparisons. Exact *P* values for significant genes were: TNFRSF9, $P = 0.003568$; LAG3, $P = 0.001110$; CD86, $P = 0.000966$; HHLA2, $P = 0.021844$; CD48, $P = 3.761 \times 10^{-5}$; TNFSF4, $P = 0.017260$; CD274, $P = 0.005823$; CD80, $P = 0.002208$; CD244, $P = 0.000297$; CD70, $P = 0.003349$; IDO1, $P = 0.000032$; HAVCR2, $P = 0.008902$; ICOSLG, $P = 0.007485$; CTLA4, $P = 0.000379$; ICOS, $P = 0.000063$; LAIR1, $P = 0.015100$; KIR3DL1, $P = 0.000102$; TIGIT, $P = 0.000160$; BTLA, $P = 0.002725$; TNFRSF4, $P = 0.014303$; PDCD1, $P = 0.000275$; IDO2, $P = 0.036244$; and PDCD1LG2, $P = 0.000021$. **H** Violin plots comparing, from left to right, the IFN-γ/effector T-cell, IFN-γ 18-gene, IFN-γ 6-gene and effector T-cell signature scores between the high-risk ($n = 28$) and low-risk ($n = 54$) groups. Each data point represents one independent patient and biological replicate. The violin outlines show the kernel-density distribution; the internal boxes span the 25th–75th percentiles, centre lines indicate medians, and whiskers extend to the most extreme values within 1.5 times the interquartile range. Groups were compared using two-sided Mann–Whitney *U* tests without adjustment for multiple comparisons. Exact *P* values were $P = 4.276 \times 10^{-10}$ for the IFN-γ/effector T-cell signature, $P = 2.522 \times 10^{-9}$ for the IFN-γ 18-gene signature, $P = 1.226 \times 10^{-8}$ for the IFN-γ 6-gene signature and $P = 2.375 \times 10^{-9}$ for the effector T-cell signature. No technical replicate was treated as an independent observation. *$P < 0.05$, **$P < 0.01$ and ***$P < 0.001$. CE cancer ecotype; DEG differentially expressed gene; GSEA gene set enrichment analysis; GSVA gene set variation analysis; IFN interferon; RCSA Radiopathomics-Clinical Stratification Assessment. Source data are provided as a Source Data file.

Supplementary Methods. A two-sided *P* value < 0.05 was considered statistically significant.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

This study used CT and pathology HE images from the TCGA-STAD collection in The Cancer Imaging Archive (https://www.cancerimagingarchive.net/collection/tcga-stad/). The RNA-sequencing data produced in this work have been archived in the Genome Sequence Archive for Human (GSA-Human) at the National Genomics Data Center (NGDC) and can be retrieved under accession HRA013404. Because these sequencing data carry a risk of patient re-identification, they are held under managed access in keeping with the applicable privacy regulations; investigators who wish to reuse them should apply to the GSA-Human Data Access Committee and provide a short outline of the planned study, the specific data fields needed, and the intended analyses, together with the committee's standard request form. Individual-level clinical records, along with the in-house CT and HE images, cannot be shared openly because they are governed by national privacy legislation and by the conditions of our institutional review board. Researchers seeking these materials should send a written project proposal to the corresponding author, Qun Zhao (zhaoqun@hebmu.edu.cn). Each application is assessed by the institutional data-governance committee on the basis of regulatory compliance and scientific rationale, with a decision returned within two weeks; once a request is approved, the data remain available for the agreed lifetime of that project. Processed sequencing results can likewise be obtained from GSA-Human. Source data are provided with this paper.

Code availability

All code used for the computational analyses in this study is available at (https://github.com/hebeidpa/GC-MLM-RCSA), and the exact version used for this study is archived on *Zenodo* (DOI: 10.5281/zenodo.20667762)[46]. The code is released under the MIT License, an Open Source Initiative-approved license, and may be reused without restriction under the terms of that license.

References

1. Siegel, R. L., Kratzer, T. B., Giaquinto, A. N., Sung, H. & Jemal, A. Cancer statistics, 2025. *CA Cancer J. Clin.* **75**, 10–45 (2025).
2. Bray, F. et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J. Clin.* **74**, 229–263 (2024).
3. Joshi, S. S. & Badgwell, B. D. Current treatment and recent progress in gastric cancer. *CA Cancer J. Clin.* **71**, 264–279 (2021).
4. Sexton, R. E., Al Hallak, M. N., Diab, M. & Azmi, A. S. Gastric cancer: a comprehensive review of current and future treatment strategies. *Cancer Metas. Rev.* **39**, 1179–1203 (2020).
5. Takahashi, K. et al. Surgical treatment for liver metastasis from gastric cancer: a systematic review and meta-analysis of long-term outcomes and prognostic factors. *Eur. J. Surg. Oncol.* **50**, 108582 (2024).
6. Conde Monroy, D. et al. Survival outcomes of hepatectomy in gastric cancer liver metastasis: a systematic review and meta-analysis. *J. Clin. Med.* **12**, 704 (2023).
7. Wang, C., Zhang, Y., Zhang, Y. & Li, B. A bibliometric analysis of gastric cancer liver metastases: advances in mechanisms of occurrence and treatment options. *Int J. Surg.* **110**, 2288–2299 (2024).
8. Li, G. Z., Doherty, G. M. & Wang, J. Surgical management of gastric cancer: a review. *JAMA Surg.* **157**, 446–454 (2022).

9. Spolverato, G. et al. Development of a prognostic nomogram and nomogram software application tool to predict overall survival and disease-free survival after curative-intent gastrectomy for gastric cancer. *Ann. Surg. Oncol.* **29**, 1220–1229 (2022).
10. Ding, P. et al. Transcriptomics-based liquid biopsy for early detection of recurrence in locally advanced gastric cancer. *Adv. Sci. (Weinh.).* **11**, e2406276 (2024).
11. Chen, Q. et al. Radiomics in precision medicine for gastric cancer: opportunities and challenges. *Eur. Radiol.* **32**, 5852–5868 (2022).
12. Wang, W. et al. Development and validation of a computed tomography-based radiomics signature to predict response to neoadjuvant chemotherapy for locally advanced gastric cancer. *JAMA Netw. Open.* **4**, e2121143 (2021).
13. Sun, Z. et al. Comprehensive assessment of immune context and immunotherapy response via noninvasive imaging in gastric cancer. *J. Clin. Invest.* **134**, e175834 (2024).
14. Sun, Z. et al. Noninvasive imaging evaluation of peritoneal recurrence and chemotherapy benefit in gastric cancer after gastrectomy: a multicenter study. *Int J. Surg.* **109**, 2010–2024 (2023).
15. Adili, D., Mohetaer, A. & Zhang, W. Diagnostic accuracy of radiomics-based machine learning for neoadjuvant chemotherapy response and survival prediction in gastric cancer patients: A systematic review and meta-analysis. *Eur. J. Radiol.* **173**, 111249 (2024).
16. Shen, L. L. et al. Delta computed tomography radiomics features-based nomogram predicts long-term efficacy after neoadjuvant chemotherapy in advanced gastric cancer. *Radio. Med.* **128**, 402–414 (2023).
17. Wang, X. et al. A pathology foundation model for cancer diagnosis and prognosis prediction. *Nature* **634**, 970–978 (2024).
18. Chang, L. et al. Advancing precision medicine: the transformative role of artificial intelligence in immunogenomics, radiomics, and pathomics for biomarker discovery and immunotherapy optimization. *Cancer Biol. Med.* **22**, 33–47 (2025).
19. Prelaj, A. et al. Artificial intelligence for predictive biomarker discovery in immuno-oncology: a systematic review. *Ann. Oncol.* **35**, 29–65 (2024).
20. Zhang, Y. et al. Histopathology images-based deep learning prediction of prognosis and therapeutic response in small cell lung cancer. *NPJ Digit Med.* **7**, 15 (2024).
21. Han, Z. et al. Development and interpretation of a pathomics-driven ensemble model for predicting the response to immunotherapy in gastric cancer. *J. Immunother. Cancer* **12**, e008927 (2024).
22. Wang, R. et al. Development of a novel combined nomogram model integrating deep learning-pathomics, radiomics and immunoscore to predict postoperative outcome of colorectal cancer lung metastasis patients. *J. Hematol. Oncol.* **15**, 11 (2022).
23. Jiang, W. et al. Pathomics signature for prognosis and chemotherapy benefits in stage III colon cancer. *JAMA Surg.* **159**, 519–528 (2024).
24. Yuan, W. et al. Pancancer outcome prediction via a unified weakly supervised deep learning model. *Signal Transduct. Target Ther.* **10**, 285 (2025).
25. Wang, H. et al. Deep learning-based pathomics signature predicts prognosis and treatment response in gastric cancer: a multicenter retrospective study. *NPJ Precis Oncol.* **10**, 206 (2026).
26. Chen, D. et al. Prognostic and predictive value of a pathomics signature in gastric cancer. *Nat. Commun.* **13**, 6903 (2022).
27. Tian, Y. et al. Neoadjuvant docetaxel, oxaliplatin plus capecitabine versus oxaliplatin plus capecitabine for patients with locally advanced gastric adenocarcinoma: long-term results of a phase III randomized controlled trial. *Int J. Surg.* **109**, 4000–4008 (2023).
28. Cao, M. et al. Development and validation of a deep learning model for predicting gastric cancer recurrence based on CT imaging: a multicenter study. *Int J. Surg.* **110**, 7598–7606 (2024).
29. Chen, D. et al. Predicting peritoneal recurrence in gastric cancer with serosal invasion using a pathomics nomogram. *iScience* **26**, 106246 (2023).
30. Cui, Y. et al. A CT-based deep learning radiomics nomogram for predicting the response to neoadjuvant chemotherapy in patients with locally advanced gastric cancer: a multicenter cohort study. *EClinicalMedicine* **46**, 101348 (2022).
31. Lu, Z. et al. The landscape of cancer research and cancer care in China. *Nat. Med.* **29**, 3022–3032 (2023).
32. Zeng, D. et al. Tumor microenvironment characterization in gastric cancer identifies prognostic and immunotherapeutically relevant gene signatures. *Cancer Immunol. Res.* **7**, 737–750 (2019).
33. de Visser, K. E. & Joyce, J. A. The evolving tumor microenvironment: from cancer initiation to metastatic outgrowth. *Cancer Cell.* **41**, 374–403 (2023).
34. Bagaev, A. et al. Conserved pan-cancer microenvironment subtypes predict response to immunotherapy. *Cancer Cell.* **39**, 845–865.e7 (2021).
35. Yoon, H. H. et al. Association of PD-L1 expression and other variables with benefit from immune checkpoint inhibition in advanced gastroesophageal cancer: systematic review and meta-analysis of 17 phase 3 randomized clinical trials. *JAMA Oncol.* **8**, 1456–1465 (2022).
36. Baxter, M. A. et al. An investigation of the clinical impact and therapeutic relevance of a DNA damage immune response (DDIR) signature in patients with advanced gastroesophageal adenocarcinoma. *ESMO Open.* **9**, 103450 (2024).
37. Collins, G. S. et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* **385**, e078378 (2024).
38. von Elm, E. et al. STROBE initiative. the strengthening the reporting of observational studies in epidemiology (STROBE) statement: guidelines for reporting observational studies. *Lancet* **370**, 1453–1457 (2007).
39. Wang, F. H. et al. The Chinese Society of Clinical Oncology (CSCO): clinical guidelines for the diagnosis and treatment of gastric cancer, 2023. *Cancer Commun. (Lond.).* **44**, 127–172 (2024).
40. Ajani, J. A. et al. Gastric cancer, version 2.2022, NCCN clinical practice guidelines in oncology. *J. Natl. Compr. Canc Netw.* **20**, 167–192 (2022).
41. Chen, R. J. et al. Towards a general-purpose foundation model for computational pathology. *Nat. Med.* **30**, 850–862 (2024).
42. van Buuren, S. Flexible Imputation of Missing Data. Second Edition (2nd ed.). Chapman and Hall/CRC. (2018).
43. Riley, R. D. et al. Calculating the sample size required for developing a clinical prediction model. *BMJ* **368**, m441 (2020).
44. Vaswani, A. et al. Attention is all you need. Preprint at *arXiv* https://doi.org/10.48550/arXiv.1706.03762 (2017).
45. Carson, J. L. et al. MINT investigators. restrictive or liberal transfusion strategy in myocardial infarction and anemia. *N. Engl. J. Med.* **389**, 2446–2456 (2023).
46. hebeidpa. hebeidpa/GC-MLM-RCSA: V1.1. Version v1.1, *Zenodo* https://doi.org/10.5281/zenodo.20667762 (2026).

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Author contributions

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Correspondence and requests for materials should be addressed to Ping'an Ding, Sen Yang, Lingjiao Meng or Qun Zhao.

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